

Gerald Mahapranaja Pillian

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Surabaya, East Java, Indonesia — Open to international relocation

Summary

Information Systems graduate of Institut Teknologi Sepuluh Nopember (ITS), Indonesia, with a cross-disciplinary background in data analytics, machine learning, UI/UX design, and intelligent systems. Completed undergraduate thesis in educational data mining, published research on mobile security, and participated in international exchange at NTUST Taiwan. Seeking a Master's programme in Data Science, Information Systems, or a related field to deepen expertise in applied machine learning and human-centred computing.

Education

- Bachelor of Information Systems (S.Kom.)** Aug 2022 – Sep 2026
Institut Teknologi Sepuluh Nopember (ITS), Faculty of Intelligent Electrical & Informatics Technology Surabaya, Indonesia
– Cumulative GPA: **3.51 / 4.00**
– Undergraduate Thesis: *Hierarchical Cluster Analysis of Medical Student Performance Across Multiple Assessment Modalities* – applied Ward's linkage hierarchical clustering ($k=3$) to 98 students' multimodal assessment data; Silhouette Coefficient 0.3913, CH Index 61.70; supervised by Ahmad Muklason, S.Kom., M.Sc., Ph.D.
– Awards: Runner-up FTEIC 2022 Business Plan Competition; Semi-Finalist UGM BCC (2023); Semi-Finalist Danamon DIBPC (2024)
- Exchange Student – Department of Computer Science** Aug 2025 – Jan 2026
National Taiwan University of Science and Technology (NTUST) Taipei, Taiwan
– Competitively selected institutional exchange programme
– Completed projects in machine learning, fuzzy logic control systems, and network communication simulation
- Secondary Education – Natural Sciences Track** Oct 2019 – Oct 2022
Lazuardi High School Indonesia

Research & Publications

- Hierarchical Cluster Analysis of Medical Student Performance Across Multiple Assessment Modalities** 2025
Jurnal Teknik ITS, co-authored with Ahmad Muklason, S.Kom., M.Sc., Ph.D. (ITS)
– Identified three performance clusters – At-Risk (26.5%), Mixed Performance (9.2%), Average (64.3%) – from 98 medical students across written, practical, standardised patient, and problem-based assessment modalities
– Validated via one-way ANOVA ($p < 0.001$), Silhouette Coefficient, Calinski-Harabasz, and Davies-Bouldin indices; findings inform early academic intervention frameworks in medical education
- Understanding User Perceptions of Camera and Microphone Access in Mobile Applications** 2024
ResearchGate
– Investigated user trust patterns and security/privacy implications of application permission systems on mobile platforms

Notable Projects

- Drone-Based Pipeline Visual Inspection using YOLOv11n with GPS Localization** 2025
Collaborative Research – Marine Engineering, ITS
– Contributed to a cross-departmental research project developing a drone-assisted inspection framework for oil, gas, and maritime pipeline infrastructure
– System integrates YOLOv11n lightweight object detection, TensorFlow, and GPS-based geotagging to identify corrosion and third-party damage on surface pipelines in real time
– Dataset prepared through field-acquired imagery, public corrosion datasets, and augmentation pipelines; model evaluated on precision, recall, F1-score, mAP, and inference time
– GPS integration enables spatial defect mapping to support predictive maintenance planning and reduce reliance on manual inspection methods
- Particle Swarm Optimised Neural Networks for Air Quality Forecasting** 2025
NTUST – Machine Learning Course
– Built and trained a neural network on 100,000+ samples from Taipei urban sensor data; used PSO for hyperparameter optimisation to forecast real-time PM2.5 and AQI levels
- Fuzzy Logic Control for Two-Axis Autonomous Aiming Systems** 2025
NTUST
– Designed a fuzzy logic controller for real-time dual-axis target tracking; implemented and tested in CoppeliaSim and MATLAB simulation environments

- CLASTIC – AI-Based Plastic Waste Sorting Mobile Application** 2024
Data Engineering & BI Lab, ITS

- Led UI/UX design for an AI-powered waste management application deployed in rural Indonesia; integrated ML-based plastic classification with field usability testing

Email Communication Simulation System

2025

NTUST

- Engineered a simulation of SMTP/IMAP email communication protocols; focused on server-client architecture and protocol flow modelling

Professional Experience & Involvement

UI/UX Designer & Developer

May 2024 – Jun 2024

Data Engineering and Business Intelligence Laboratory, IS ITS

Surabaya, Indonesia

- Designed end-to-end UI/UX for CLASTIC, an AI-powered plastic waste sorting mobile application; conducted field testing in Sumberbening Village
- Produced complete design system covering UI components, web assets, and graphic design

Expert Staff – External Relations

Apr 2023 – Aug 2025

Information Systems Expo (ISE!), ITS

Surabaya, Indonesia

- Progressed from General Staff to Expert Staff; managed and mentored a team of 3 staff members
- Served as Master of Ceremonies for UX Academy and Data Software Academy; achieved KPI of 300+ participants

Student Staff Volunteer – Global Engagement Office

Dec 2023 – Feb 2024

Institut Teknologi Sepuluh Nopember

Surabaya, Indonesia

- Liaison for 100+ international students across 10+ major events; roles included speaker, MC, logistics coordinator, and campus guide

Assistant Lecturer – Introduction to Information Systems

Sep 2023 – Dec 2023

Department of Information Systems, ITS

Surabaya, Indonesia

- Designed assignments and projects; provided academic support and grading for undergraduate cohort

Strategy & Development Staff

Jun 2023 – Feb 2024

IEEE ITS Surabaya Student Branch

Surabaya, Indonesia

- Principal editor for marketing and archival content; completed 5+ major projects on the Documentation and Editorial Team

Certifications

SAP Certified Application Associate

Jul 2024

SAP BusinessObjects BI Platform 4.2 — SAP University Alliances

Online

- 3-month ERP programme covering SD, MM, and PP modules

Data Analyst Certification

Aug – Oct 2023

MySkill

Online

- Hands-on training in SQL, Python, and DBMS; completed guided capstone data analysis project

Technical Skills

Programming & Query: Python (NumPy, pandas, scikit-learn, scipy, matplotlib, TensorFlow), SQL, R, JavaScript, PHP/Laravel, Java, VBA, HTML/CSS

Machine Learning & Data: Supervised/Unsupervised Learning, Neural Networks, Object Detection (YOLOv11n), Particle Swarm Optimisation, Hierarchical Clustering, SAP ERP & BI Platform, Power BI, R Studio

Simulation & Control Systems: MATLAB, CoppeliaSim, Fuzzy Logic Control, GPS-based Localization

Web & App Development: Supabase, Google Cloud APIs, REST API Integration, Laravel

Design & Prototyping: Figma (UI/UX), Video Editing, Professional Photography, Digital Marketing

Tools & Platforms: Git/GitHub, Microsoft Office (Advanced), Google Workspace, GDevelop, L^AT_EX/Overleaf

Languages

Indonesian – Native (C2) **English** – Fluent (C1) **Javanese** – Intermediate (B1) **Mandarin Chinese** – Basic (A2)

Arabic – Basic (A2) **French** – Basic (A1) **German** – Basic (A1)

Additional Information

Nationality: Indonesian **Availability:** From June 2026

Research Interests: Educational data mining, applied machine learning, computer vision, intelligent systems, human-computer interaction

References: Available upon request (academic supervisor: Ahmad Muklason, S.Kom., M.Sc., Ph.D., ITS)